

Implicit assessment in psychology: Myths, doubts, and practical applications

Dr. Ottavia M. Epifania
ottavia.epifania@unipd.it

University of Padova

June 9th-10th 2022

Contents

- 1 Implicit assessment: Definitions and evolution
- 2 The Implicit Association Test
- 3 What's wrong with the IAT
- 4 One, not the only one
- 5 Together is not always better
- 6 A possible workaround
- 7 Rasch gone mixed
- 8 The easy way

IMPLICIT ASSESSMENT

In Greenwald & Banaji (1995), implicit attitudes are taken to be:

Introspectively unidentified—or inaccurately identified—traces of past experience that mediate favorable or unfavorable feelings, thoughts, or actions toward social objects

IMPLICIT = UNCONSCIOUS

Implicit attitudes express themselves through **automatic associations**

Greenwald, A. G., & Banaji, M. R. (1995) Implicit Social Cognition: Attitudes, Self-Esteem, and Stereotypes. *Psychological Review*, 102-1, doi: 10.1037/0033-295X.102.1.4

Automatic associations



- Beyond one's control
- Activated by triggering stimuli
- Not subject to introspection
- Fast and almost immediate

Automatic associations and the unconscious

Studying automatic associations = studying implicit attitudes



Access the unconscious and investigate it

Automatic associations and the unconscious

Studying automatic associations = studying implicit attitudes



Access the unconscious and investigate it

WRONG!

Fazio & Olson (2003):

Being fast is associating snakes with negative attributes does not imply that you are unaware of your negative attitudes towards snakes!

The only thing you are unaware of: Your attitudes are being assessed → something is indeed implicit

Implicitly assessed constructs vs. unconscious constructs

Fazio, R. H., & Olson, M. A. (2003). Implicit measures in social cognition research: Their meaning and use. *Annual Review of Psychology*, 54(1), 297– 327.

<https://doi.org/10.1146/annurev.psych.54.101.601.145225>

From:

Explicit = conscious Vs. Implicit = unconscious

to:

Explicit = direct Vs. Implicit = indirect

Empirical meaning of implicit

Banaji & Greenwald (2013):

Theoretical definition of implicit as unconscious

Greenwald & Banaji (2017), Greenwald & Lai (2020):

Empirical definition of implicit as indirect

Banaji, M. R., & Greenwald, A. G. (2013). *Blindspot: Hidden biases of good people*. Delacorte Press,

Greenwald, A. G., & Banaji, M. R. (2017). The implicit revolution: Reconceiving the relation between conscious and unconscious. *American Psychologist*, 72(9), 861–871. <https://doi.org/10.1037/amp0000238>

Greenwald, A. G., & Lai, C. K. (2020). Implicit social cognition. *Annual Review of Psychology*, 71, 419–445. <https://doi.org/10.1146/annurev-psych-010419-050837>

Why...?

- Lack of evidence for supporting the unconsciousness
- Issues with construct validity
- Theory uncommitted definition

IAT

Implicit Association Test

Greenwald et al. (1998):



Table: IAT

Block	Trial	Left key	Right key
1	20	Good	Bad
2	20	Coke	Pepsi
3	20	Coke + Good	Pepsi + Bad
4	40	Coke + Good	Pepsi + Bad
5	20	Pepsi	Coke
6	20	Pepsi + Good	Coke + Bad
7	40	Pepsi + Good	Coke + Bad

Implicit Association Test

Greenwald et al. (1998):



Table: IAT

Block	Trial	Left key	Right key
1	20	Good	Bad
2	20	Coke	Pepsi
3	20	Coke + Good	Pepsi + Bad
4	40	Coke + Good	Pepsi + Bad
5	20	Pepsi	Coke
6	20	Pepsi + Good	Coke + Bad
7	40	Pepsi + Good	Coke + Bad

Coke-Good/Pepsi-Bad Condition

Implicit Association Test

Greenwald et al. (1998):



Table: IAT

Block	Trial	Left key	Right key
1	20	Good	Bad
2	20	Coke	Pepsi
3	20	Coke + Good	Pepsi + Bad
4	40	Coke + Good	Pepsi + Bad
5	20	Pepsi	Coke
6	20	Pepsi + Good	Coke + Bad
7	40	Pepsi + Good	Coke + Bad

Coke-Good/Pepsi-Bad Condition

Pepsi-Bad/Coke-Good Condition

Coke-Good/Pepsi-Bad (CGPB)

Coke
Good



Pepsi
Bad

Pepsi-Good/Coke-Bad (PGCB)

Pepsi
Good



Coke
Bad

D score

Greenwald et al. (2003)

$$D_{B6,B3} = \frac{M_{B6} - M_{B4}}{sd_{B6,B3}}$$

$$D_{B7,B4} = \frac{M_{B7} - M_{B4}}{sd_{B7,B4}}$$

$$D = \frac{D_{B6,B3} + D_{B7,B4}}{2}$$

Error responses? Fast responses?

Greenwald, A. G., Nosek, B. A., & Banaji, M. R. (2003). Understanding and using the implicit association test: I. An improved scoring algorithm. *Journal of Personality and Social Psychology*, 85(2), 197– 216. <https://doi.org/10.1037/0022-3514.85.2.197>

Table: Overview of the D score algorithms.

D score	Error inflation	Lower tail treatment
$D1$	Built-in correction	No
$D2$	Built-in correction	Delete trials < 400 ms
$D3$	Mean (correct responses) $+ 2sd$	No
$D4$	Mean (correct responses) $+ 600$ ms	No
$D5$	Mean (correct responses) $+ 2sd$	Delete trials < 400 ms
$D6$	Mean (correct responses) $+ 600$ ms	Delete trials < 400 ms

WHAT'S WRONG

A measure in search of a construct

- Lack of correlation with explicit measures
- Lack of correlation with actual behaviors
- Results are hard to reproduce

YES BUT

- Correlation with explicit measures in socially non relevant topics
- The type of behavior one tries to predict

Carlsson, R., & Agerstrom, J. (2016). A closer look at the discrimination outcomes in the IAT literature. *Scandinavian Journal of Psychology*, 57(4), 278–287. doi:

<https://doi.org/10.1111/sjop.12288>

Schimmack, U. (2021). The Implicit Association Test: A method in search of a construct. *Perspectives on Psychological Science*, 16(2), 396–414. <https://doi.org/10.1177/1745691619863798>

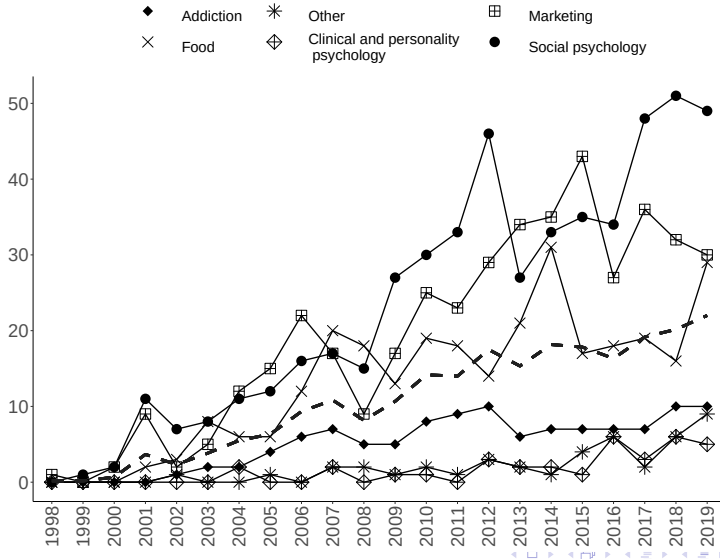
Laziness

- What algorithm did you use? Don't know, don't care
- Error prone procedure
- Lack of automated and free software options

The D score

	Condition A	Condition B
Francesco	  	  
Giulia	  	  
Jessica	  	  

However...



NOT THE ONLY ONE



VS



**VS**

IAT → Comparative measure between contrasting objects:

- The contrasted category is not always clear
- The measure of the target category depends on the contrasted category
- Maybe you want an “absolute” measure....



Single Category IAT

Karpinski & Steinman (2006):

Table: SC-IAT

Block	Trial	Left key	Right key
1	24	Good + Coke	Bad
2	72	Good + Coke	Bad
3	24	Good	Bad + Coke
4	72	Good	Bad + Coke



Single Category IAT

Karpinski & Steinman (2006):

Table: SC-IAT

Block	Trial	Left key	Right key
1	24	Good + Coke	Bad
2	72	Good + Coke	Bad
3	24	Good	Bad + Coke
4	72	Good	Bad + Coke

Coke-Good condition



Single Category IAT

Karpinski & Steinman (2006):

Table: SC-IAT

Block	Trial	Left key	Right key
1	24	Good + Coke	Bad
2	72	Good + Coke	Bad
3	24	Good	Bad + Coke
4	72	Good	Bad + Coke

Coke-Good condition

Coke-Bad condition



Single Category IAT

Karpinski & Steinman (2006):

Table: SC-IAT

Block	Trial	Left key	Right key
1	24	Good + Coke	Bad
2	72	Good + Coke	Bad
3	24	Good	Bad + Coke
4	72	Good	Bad + Coke

Coke-Good condition

Coke-Bad condition

- Response Time Window (rtw, usually 1,500 ms)
- Feedback to each response

Scoring the SC-IAT

$$D_{score} = \frac{M_{B2} - M_{B4}}{S_{B2,B4}}$$

- Responses < 350ms → discarded
- Responses > *rtw* → discarded
- Error responses → $M + 400\text{ms}$

Karpinski, A., & Steinman, R. B. (2006). The single category implicit association test as a measure of implicit social cognition. *Journal of Personality and Social Psychology*, 91(1), 16–32. <https://doi.org/10.1037/0022-3514.91.1.16>

Go/No-go association Task

Nosek & Banaji (2001):

Table: GNAT

Block	Trials	Signal	Noise
1	16	Good	Bad
2	16	Bad	Good
3	16	Fruit	Bugs
4	16	Bugs	Fruit
5	16	Good + Fruit	Bad + Bugs
	60	Good + Fruit	Bad + Bugs
6	16	Bad + Fruit	Good + Bugs
	60	Bad + Fruit	Good + Bugs
7	16	Good + Bugs	Bad + Fruit
	60	Good + Bugs	Bad + Fruit
8	16	Bad + Bugs	Good + Fruit
	60	Bad + Bugs	Good + Fruit

Go/No-go association Task

Nosek & Banaji (2001):

Table: GNAT

Block	Trials	Signal	Noise
1	16	Good	Bad
2	16	Bad	Good
3	16	Fruit	Bugs
4	16	Bugs	Fruit
5	16	Good + Fruit	Bad + Bugs
	60	Good + Fruit	Bad + Bugs
6	16	Bad + Fruit	Good + Bugs
	60	Bad + Fruit	Good + Bugs
7	16	Good + Bugs	Bad + Fruit
	60	Good + Bugs	Bad + Fruit
8	16	Bad + Bugs	Good + Fruit
	60	Bad + Bugs	Good + Fruit

Go/No-go association Task

Nosek & Banaji (2001):

Table: GNAT

Block	Trials	Signal	Noise
1	16	Good	Bad
2	16	Bad	Good
3	16	Fruit	Bugs
4	16	Bugs	Fruit
5	16	Good + Fruit	Bad + Bugs
	60	Good + Fruit	Bad + Bugs
6	16	Bad + Fruit	Good + Bugs
	60	Bad + Fruit	Good + Bugs
7	16	Good + Bugs	Bad + Fruit
	60	Good + Bugs	Bad + Fruit
8	16	Bad + Bugs	Good + Fruit
	60	Bad + Bugs	Good + Fruit

GO

Fruit



Good

NO GO

Fruit



Good

Nosek, B. A., & Banaji, M. R. (2001). The go/no-go association task. *Social Cognition*, 19(6), 625–666. Q21 <https://doi.org/10.1521/soco.19.6.625.20886>

Scoring

d' (Green & Swets, 1966):

- ① z score of the proportion of hits (correct “go” responses for signal items)
- ② z score of the proportion of false alarms (incorrect “go” responses for noise items)
- ③ Difference between the two scores

Empty cells (no false alarms or misses) $\rightarrow 0.35/\text{number of trials}$

Sorting Paired Features Task

Bar-Anan et. al (2009):

Table: SPF

Blocks	Trials	Top-left	Top-right	Bottom-left	Bottom-right
1	48	Dogs+Good	Dogs+Bad	Cats+Good	Cats+Bad
2	48	Dogs+Bad	Dogs+Good	Cats+Bad	Cats+Good
3	48	Cats+Bad	Cats+Good	Dogs+Bad	Dogs+Good
4	48	Cats+Good	Cats+Bad	Dogs+Good	Dogs+Bad

To each pairing → a response key

Bar-Anan, Y., Nosek, B. A., & Vianello, M. (2009). The sorting paired features task: A measure of association strengths. *Experimental Psychology*, 56(5), 329–343.

<https://doi.org/10.1027/1618-3169.56.5.329>

Correct key: M

Dogs Bad Key: Q	Lovely	Dogs Good Key: P
		
Cats Good Key: C		Cats Good Key: M

Correct key: P

Dogs Bad Key: Q	Lovely	Dogs Good Key: P
		
Cats Good Key: C		Cats Good Key: M

TOGETHER



VS



IAT



SC-IAT 1



SC-IAT 2



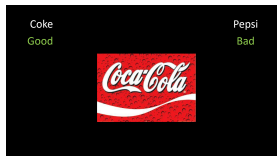
IAT



SC-IAT 1



SC-IAT 2

 D score D Coke D Pepsi

Stand alone measure:

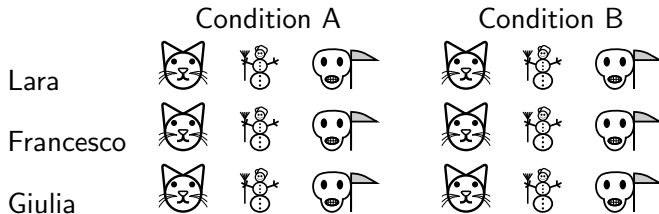
Multiple measures:

Stand alone measure:

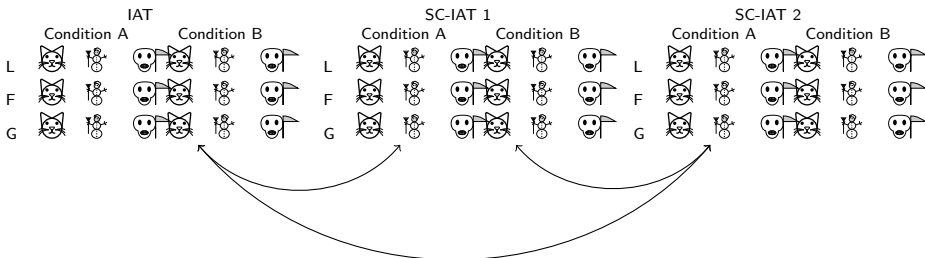
	Condition A			Condition B		
Lara						
Francesco						
Giulia						

Multiple measures:

Stand alone measure:



Multiple measures:



THE WORKAROUND

The supermodel(s)

Linear Mixed-Effects Models (LMMs)



Account for (potentially) all the sources of variability and dependency



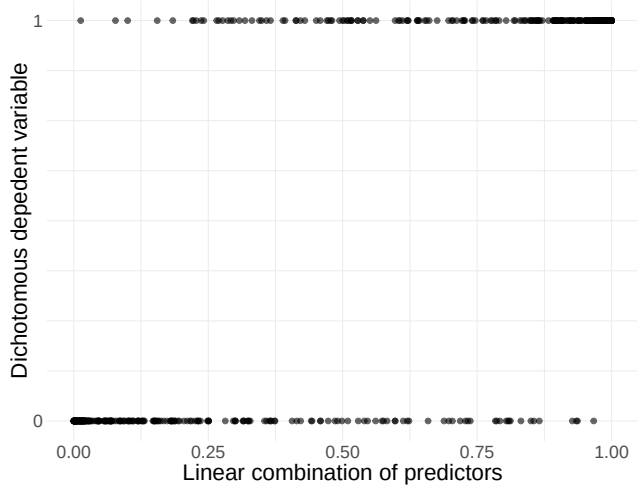
Gather information at the stimulus level



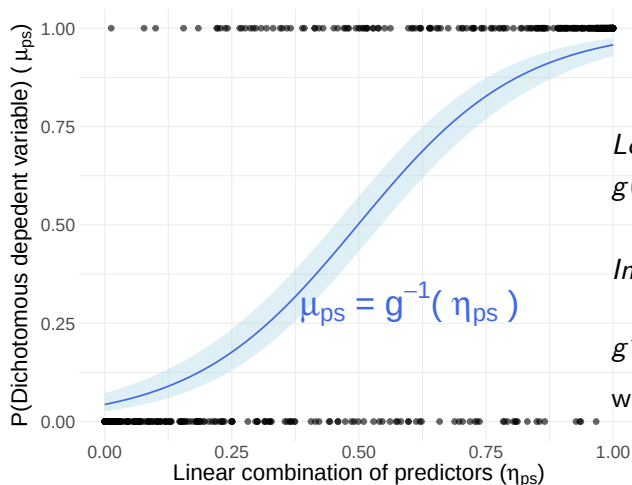
Estimate the Rasch and log-normal models

Epifania, O. M., Robusto, E., & Anselmi, P. (2021). Rasch gone mixed: A mixed model approach to the Implicit Association Test. *Testing, Psychometrics, Methodology in Applied Psychology*, 28, 467–483. doi: 10.4473/TPM28.4.5

From Linear to Rasch and log-normal



From Linear to Rasch and log-normal



Logit link function g
 $g(\eta_{ps}) = \log \left(\frac{\mu_{ps}}{1-\mu_{ps}} \right)$

Inverse g^{-1}

$$g^{-1} = \frac{\exp(\eta_{ps})}{1 + \exp(\eta_{ps})}$$

where $\eta_{ps} = \theta_p + b_s$

Standard

(G)LM

Rasch model

$$P(x_{ps} = 1) = \frac{\exp(\theta_p - b_s)}{1 + \exp(\theta_p - b_s)}$$

$$P(x_{ps} = 1) = \frac{\exp(\theta_p + b_s)}{1 + \exp(\theta_p + b_s)}$$

Log-normal model

$$f(t_{ps}) = \delta_s - \tau_p + \varepsilon$$

$$f(t_{ps}) = \delta_s + \tau_p + \varepsilon$$

Standard

(G)LM

Rasch model

$$P(x_{ps} = 1) = \frac{\exp(\theta_p - b_s)}{1 + \exp(\theta_p - b_s)}$$

$$P(x_{ps} = 1) = \frac{\exp(\theta_p + b_s)}{1 + \exp(\theta_p + b_s)}$$

Log-normal model

$$f(t_{ps}) = \delta_s - \tau_p + \varepsilon$$

$$f(t_{ps}) = \delta_s + \tau_p + \varepsilon$$

The expected response y for the observation $i = 1, \dots, I$ for respondent $p = 1, \dots, P$ on stimulus $s = 1, \dots, S$ in condition $c = 1, \dots, C$:

Model 1:

$$y_i = \alpha + \beta_c X_c + \alpha_{p[i]} + \alpha_{s[i]} + \varepsilon_i$$

$$\alpha_p \sim \mathcal{N}(0, \sigma_p^2),$$

$$\alpha_s \sim \mathcal{N}(0, \sigma_s^2).$$

Fixed Effects

Model 2:

$$y_i = \alpha + \beta_c X_c + \alpha_{p[i]} + \beta_{s[i]} c_i + \varepsilon_i$$

$$\alpha_p \sim \mathcal{N}(0, \sigma_p^2),$$

$$\beta_s \sim \mathcal{MVN}(0, \Sigma_{sc}).$$

Random structure

Model 3:

$$y_i = \alpha + \beta_c X_c + \alpha_{s[i]} + \beta_{p[i]} c_i + \varepsilon_i$$

$$\alpha_s \sim \mathcal{N}(0, \sigma_s^2),$$

$$\beta_p \sim \mathcal{MVN}(0, \Sigma_{pc}).$$

Accuracy: $\epsilon \sim \text{Logistic}(0, \sigma^2)$

Log-time: $\epsilon \sim \mathcal{N}(0, \sigma^2)$

Accuracy model (Rasch Model estimates):

	Respondent	Stimulus
Model 1	Overall (θ_p)	Overall (b_s)
Model 2	Overall (θ_p)	Condition-specific (b_{sc})
Model 3	Condition-specific (θ_{pc})	Overall (b_s)

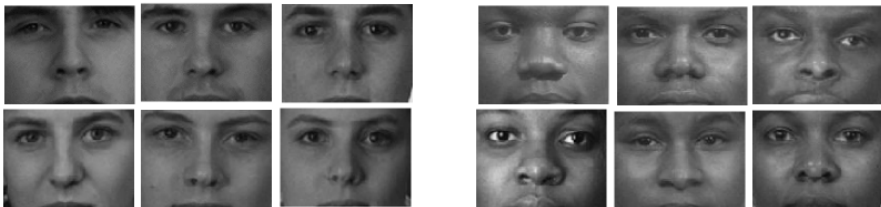
Log-time model (Log-normal Model estimates):

	Respondent	Stimulus
Model 1	Overall (τ_p)	Overall (δ_s)
Model 2	Overall (τ_p)	Condition-specific (δ_{sc})
Model 3	Condition-specific (τ_{pc})	Overall (δ_s)

RASCH GONE MIXED

Method

12 Target stimuli



16 Attributes (Good, laughter, pleasure, glory, peace, happy, joy, love and Evil, bad, horrible, terrible, nasty, pain, failure, hate)

Participants: 62 (F = 48.39%, Age = 24.92 ± 2.11 years)

Conditions:

WGBB: White-Good/Black-Bad, 60 trials

BGWB: Black-Good/White-Bad, 60 trials

Model comparison

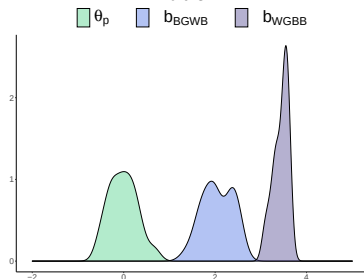
Model	Accuracy			Response times		
	AIC	BIC	Deviance	AIC	BIC	Deviance
1	3785.87	3813.53	3777.87	4762.63	4797.2	4752.63
2	3784.43	3825.91	3772.43	Aberrant estimates		
3	3786.51	3828.00	3774.51	4399.66	4448.06	4385.66

Model comparison

Model	Accuracy			Response times		
	AIC	BIC	Deviance	AIC	BIC	Deviance
1	3785.87	3813.53	3777.87	4762.63	4797.2	4752.63
2	3784.43	3825.91	3772.43	Aberrant estimates		
3	3786.51	3828.00	3774.51	4399.66	4448.06	4385.66

Rasch model:

Model 2



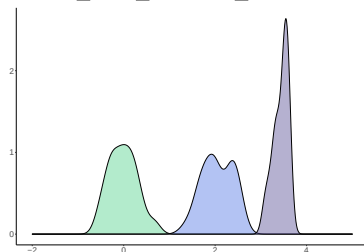
Model comparison

Model	Accuracy			Response times		
	AIC	BIC	Deviance	AIC	BIC	Deviance
1	3785.87	3813.53	3777.87	4762.63	4797.2	4752.63
2	3784.43	3825.91	3772.43	Aberrant estimates		
3	3786.51	3828.00	3774.51	4399.66	4448.06	4385.66

Rasch model:

Model 2

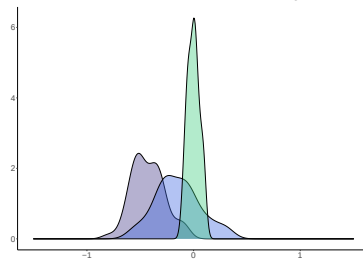
θ_p b_{BGWB} b_{WGGB}



Log-normal model:

Model 3

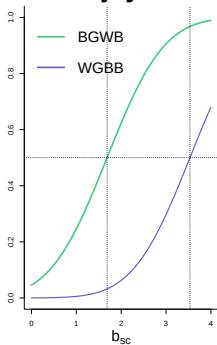
τ_{WGGB} τ_{BGWB} δ_s



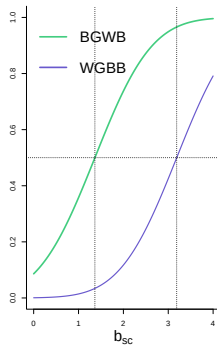
Condition-specific easiness

HIGH CONTRIBUTION STIMULI

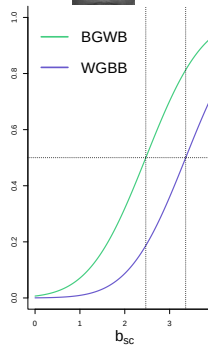
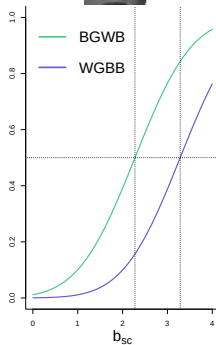
joy



evil

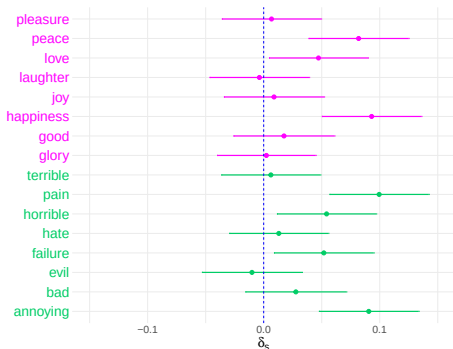


LOW CONTRIBUTION STIMULI

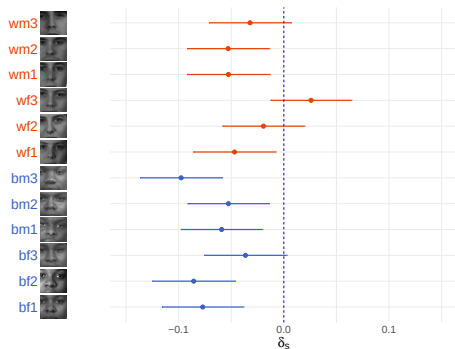


Overall time intensity

Bad & Good



Black & White



MAKE IT EASY

<i>D-score</i>	Error inflation	Delete trials < 400 ms
<i>D1</i>	Built-in correction	No
<i>D2</i>	Built-in correction	Yes
<i>D3</i>	M (correct responses) + $2sd$	No
<i>D4</i>	M (correct responses) + 600 ms	No
<i>D5</i>	M (correct responses) + $2sd$	Yes
<i>D6</i>	M (correct responses) + 600 ms	Yes

- ☹ Few Open source alternatives
- ☹ Utterly complicated to use
- ☹ No clear information on the algorithm they compute
- ☹ No graphical representation of the results

<i>D-score</i>	Error inflation	Delete trials < 400 ms
<i>D1</i>	Built-in correction	No
<i>D2</i>	Built-in correction	Yes
<i>D3</i>	M (correct responses) + $2sd$	No
<i>D4</i>	M (correct responses) + 600 ms	No
<i>D5</i>	M (correct responses) + $2sd$	Yes
<i>D6</i>	M (correct responses) + 600 ms	Yes

- ☹️ Few Open source alternatives
- ☹️ Utterly complicated to use
- ☹️ No clear information on the algorithm they compute
- ☹️ No graphical representation of the results



Something Open Source, able to compute multiple scores, and to provide nice graphical representations

Make it easy & Clear

`implicitMeasures` package DOI:10.21105/joss.02394 :

Function	Description
<code>clean_iat()</code>	Prepare and clean IAT data
<code>clean_sciat()</code>	Prepare and clean SC-IAT data
<code>compute_iat()</code>	Compute IAT <i>D</i> -score
<code>compute_sciat()</code>	Compute SC-IAT <i>D</i> -score
<code>descript_d()</code>	Descriptive table of <i>D</i> -scores (even in $\text{\LaTeX}$)
<code>d_density()</code>	Plot IAT or SC-IAT scores (distribution)
<code>d_point()</code>	Plot either IAT or SC-IAT scores (points)
<code>IAT_rel()</code>	IAT reliability
<code>multi_dsciat()</code>	Plot SC-IATs scores
<code>multi_dscore()</code>	Compute and plot multiple <i>D</i> -scores
<code>raw_data()</code>	Dataset with one IAT and two SC-IATs

- [implicitMeasures](#): Introduction to IAT, SC-IAT, and *D* scores.
- [IAT-example](#): Package illustration on IAT data
- [SC-IAT-example](#): Package illustration on SC-IAT data

Make it even easier



Source code on [GitHub](#)

DOI: [10.3389/fpsyg.2019.02938](https://doi.org/10.3389/fpsyg.2019.02938)

Thank you!

`ottavia.epifania@unipd.it`